Estimating Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

Markus Mulvihill

KTH Royal Institute of Technology

September, 2026

2026-09-07

Thesis Presentation

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2. Examiner - Ragnar Thobaben
3. Opponent - Divyayan Day

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  └─The Fundamental Problem Addressed
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   - Main Results
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Fundamentals of Glucoregulation

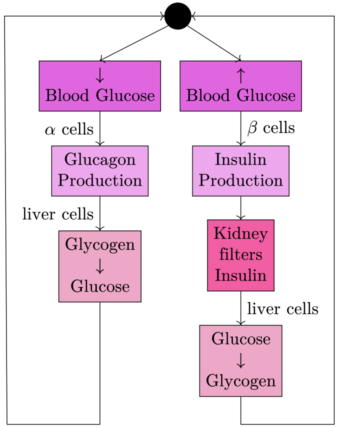
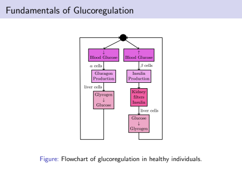


Figure: Flowchart of glucoregulation in healthy individuals.



1. The thesis revolves around the functionality of glucoregulation.
2. A high level diagram of the human insulin-glucose system can be described as the image.

Insulin Resistance in ICU Patients

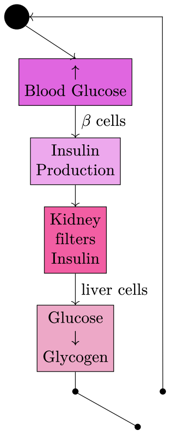
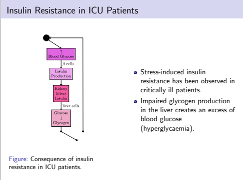


Figure: Consequence of insulin resistance in ICU patients.

- Stress-induced insulin resistance has been observed in critically ill patients.
- Impaired glycogen production in the liver creates an excess of blood glucose (hyperglycaemia).



Mention that the break in the chain is just for visual purposes.

Intensive Insulin Therapy

- **What**: Utilizes aggressive insulin infusions and feeding inputs to tightly control blood glucose.
- **Goal**: Prevents hyperglycaemia in patients to reduce mortalities and longer visitation stays.
- **How**: Traditionally has relied on ad-hoc sliding protocols.

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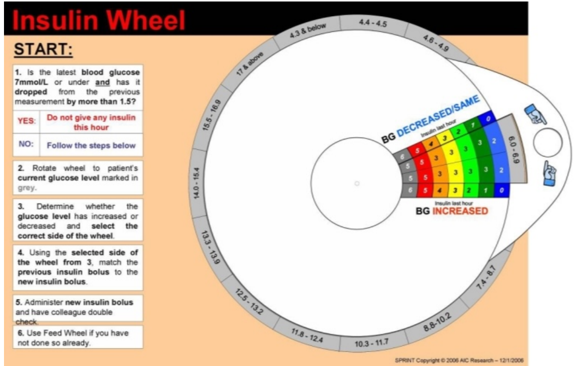

Figure: Example of the SPRINT Protocol Insulin Wheel (Lonergan et al. 2006).

- Clinicians administer insulin and feeding from complex flowcharts.
- Struggles to adapt to the dynamic health of critically ill patients.

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Overview of Ad-hoc Sliding Protocols

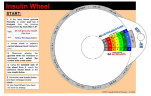

Figure: Example of the SPRINT Protocol Insulin Wheel (Lonergan et al. 2006).

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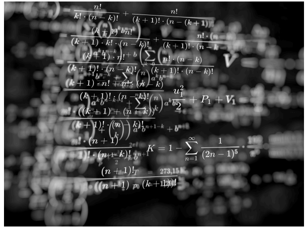
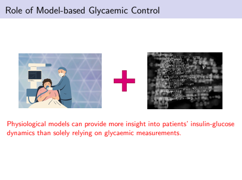
Role of Model-based Glycaemic Control



Physiological models can provide more insight into patients' insulin-glucose dynamics than solely relying on glycaemic measurements.



Insulin infusions combined with better understanding of dynamics can provide more insight to the clinicians

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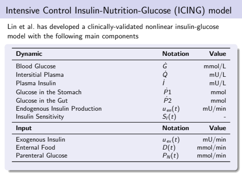
Intensive Control Insulin-Nutrition-Glucose (ICING) model

Lin et al. has developed a clinically-validated nonlinear insulin-glucose model with the following main components

Dynamic	**Notation**	**Value**
Blood Glucose	$\dot{G}$	mmol/L
Intersitial Plasma	$\dot{Q}$	mU/L
Plasma Insulin	$\dot{I}$	mU/L
Glucose in the Stomach	$\dot{P}1$	mmol
Glucose in the Gut	$\dot{P}2$	mmol
Endogenous Insulin Production	$u_{en}(t)$	mU/min
Insulin Sensitivity	$S_I(t)$	-
Input	**Notation**	**Value**
Exogenous Insulin	$u_{ex}(t)$	mU/min
Enternal Food	$D(t)$	mmol/min
Parenteral Glucose	$P_N(t)$	mmol/min



More details on the exact formulation of the dynamics can be read in the paper in further reading.

Computational Approximation of the ICING model

- The ICING model equations are defined in continuous-time
- Discretization of the model is required for the presented estimation methods

For an example dynamic F

$$
\begin{aligned}
t_k &= t_{k-1} + \Delta, \\
\dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta)}{\Delta}, \\
F_k &\approx F_{k-1} + \Delta \dot{F}.
\end{aligned}
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1. According to Euler's method
2. Discretization error is bounded by Δ

Bayesian Filtering

The discrete-time dynamic ICING model can be represented in the stochastic state-space.

State-space Representation

$$x_k \sim p(x_k \mid x_{k-1},\, u_{k-1})$$
$$y_k \sim p(y_k \mid x_k)$$

Parameter Description

System Variable	Notation
Dynamics	x_k
Inputs	u_k
Measurements	y_k

There are various methods of optimal filtering, and each has their own assumptions and limitations

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1. The measurement often depends on the hospital, but will be discussed more later
2. Just the EKF and the PF will be in the topic of this thesis

Extended Kalman Filter

Approximates nonlinear and Gaussian dynamic and measurement models as a linear transformation.

1. *Prediction step*

$$
\begin{aligned}
m_k^- &= f(m_{k-1},\, u_{k-1}),\\
P_k^- &= F_x(m_{k-1})P_{k-1}F_x^{\mathsf{T}}(m_{k-1}) + F_q(m_{k-1})Q_{k-1}F_q^{\mathsf{T}}(m_{k-1}).
\end{aligned}
$$

2. *Update step*

$$
\begin{aligned}
v_k &= y_k - h(m_k^-),\\
S_k &= H_x(m_k^-)P_k^- H_x^{\mathsf{T}}(m_k^-) + H_r(m_k^-)R_k H_r^{\mathsf{T}}(m_k^-),\\
K_k &= P_k^- H_k^{\mathsf{T}}(m_k^-)S_k^{-1},\\
m_k &= m_k^- + K_k v_k,\\
P_k &= P_k^- - K_k S_k K_k^{\mathsf{T}}.
\end{aligned}
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Bootstrap Particle Filter

Formulates the posterior of the dynamic function as a Monte Carlo approximation.

1. Sample N particles from the dynamic function

$$x_k^{(i)} \sim p(x_k \mid x_{k-1}^{(i)},\, u_{k-1}) \quad i = 1, \dots, N.$$

2. Update weights from the measurement

$$w_k^{*(i)} \propto p(y_k \mid x_k^{(i)}) \quad i = 1, \dots, N,$$
$$w_k^{(i)} = \frac{w_k^{*(i)}}{\sum_{j=1}^{N} w_k^{*(j)}} \quad i = 1, \dots, N.$$

3. Systematically resample particles from a distribution proportional to the weights

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Formulation of Synthetic Patients

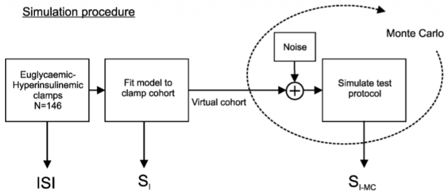

Figure: An example of a simulation procedure (Lotz et al. 2008).

- Synthetic patient trajectories model clinical insulin–glucose dynamics.
- Existing methods typically perturb previously recorded clinical data.

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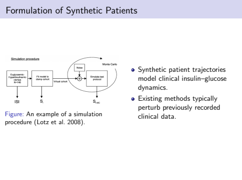

Estimation of Patient Insulin-Glucose Dynamics

- Previous studies have employed Bayesian filtering methods:
  - Extended and unscented Kalman filters
  - Particle filters
- Practical implementation is often insufficiently addressed:
  - Constant nutrition and insulin inputs
  - Known initial insulin–glucose dynamics
- Estimator configurations and performance are not systematically compared.

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Generating Synthetic Patient Trajectories

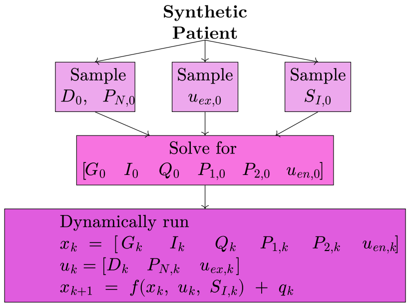
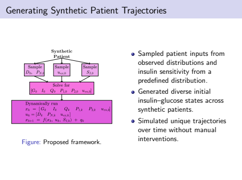


Figure: Proposed framework.

- Sampled patient inputs from observed distributions and insulin sensitivity from a predefined distribution.
- Generated diverse initial insulin–glucose states across synthetic patients.
- Simulated unique trajectories over time without manual interventions.



Synthetic Patients' Distributions

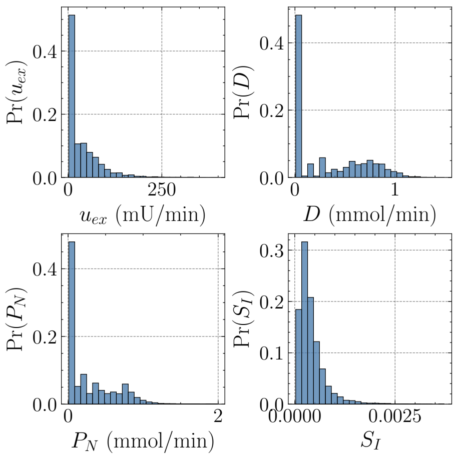


Figure: Patients' initial inputs and insulin sensitivity.

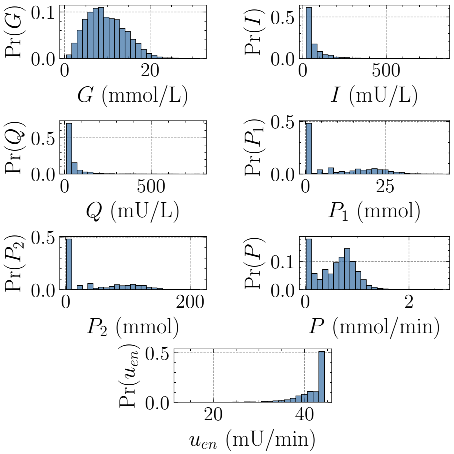
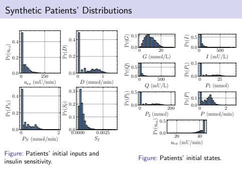


Figure: Patients' initial states.



Mention that S_I is taken as a prior from (Davidson et al. 2019)

Estimators Performance in Estimating Blood Glucose

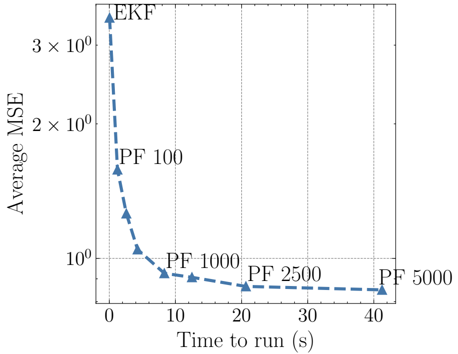
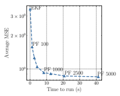


Figure: something probably

- something something

Estimators Performance in Estimating Blood Glucose



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GLUC-ICU Dataset from Karolinska University Hospital

Specifications:

1. 27113 patients in the ICU
2. 39.5% of the total number of patients given insulin at some point in their stay
3. 100 million minutes of feeding data and 80 million minutes of insulin data to sample from

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References

- Lonergan, Timothy et al. (Apr. 2006). "A Simple Insulin-Nutrition Protocol for Tight Glycemic Control in Critical Illness: Development and Protocol Comparison". In: DOI: 10.1089/dia.2006.8.191.
- Lotz, Thomas F. et al. (Mar. 2008). "Monte Carlo analysis of a new model-based method for insulin sensitivity testing". In: DOI: 10.1016/j.cmpb.2007.03.007.
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